



DEPARTMENT OF INFORMATION TECHNOLOGY

APPLICATIONS DEVELOPMENT THEORY 4 (ADTS470S)

Semester 1 Assignments

This is a group assessment which has multiple deliverables submitted in a variety of formats – hardcopy, Blackboard submissions, demonstrations or class exercises – on the due dates specified below. Please read through the following carefully:-

Project Rules regarding group work:

- ✓ Each group will consist of a maximum of 4 members.
- ✓ Students are expected to establish and manage their groups themselves. Exception cases will be mediated by the lecturer
- ✓ You may choose to do this project on your own, but remember the outcomes and expectations remain the same as if you worked in a group.
- ✓ You will need to submit part of your solution via Blackboard and your prototype will be demonstrated during class.
- ✓ All group members need to be present and contributing during the demonstration.
- ✓ A demonstration schedule will be made available during class for students to select their preferred demo slots. Any groups not meeting the schedule selection deadline will be assigned a demo slot.

The Scenario:

You are a dynamic owner of a software consultancy and wish to capitalize on the tourism boom currently being experienced. You have decided that you and your team will come up with an IT innovation for this market which stands out from the multitude of options currently available to tourists. Over the next 4 months as the team engineers the attractive solution you will produce the following:-

1. **ASSIGNMENT 1: Conceptualizing the Requirements (aka The Jam Session).**
2. **ASSIGNMENT 2: The Requirements Engineering Specification.**
3. **ASSIGNMENT 3: Design and Prototyping.**
4. **ASSIGNMENT 4: The Development.**

The Rubrics:

Assignment 1		
Due dates:-	17/2/2020	Time-boxed Class exercise

Conceptualizing the Requirements (aka The Jam Session). This will be executed during class, should consist of a single page of brainstormed results and include the problem you are addressing, a summary of your brainstorming sessions, answers to the questions in the table below, your concept, a user story, the added value and sustainability of your proposed solution.

The questions Table – please make them as **PRECISE and DETAILED** as possible.

Who are the people this the concept/idea is meant for?

What is the essence or core of the system? Remember to look beyond obvious problems to solve
 When would the concept be consumed or used?
 What is the best way or place to reach the audience?
 How will the audience gain the message or service?
 Why is this important?

TASK I: Requirements Engineering	Offered	Due date
1. The Jam Session 1 – Conceptualization (The Strategy Meeting) – in class exercise	10	17/2/2020

Assignment 2

Due dates:-	26/2/2020	Blackboard
--------------------	-----------	------------

The Requirements Engineering Specification. This should include the requirements and its proposed roadmap and release plan. This task ends with a presentation of the RE Specification in a maximum 8 slides

TASK II: Requirements Engineering	Marks	Due date
2.1 Requirements Specification Initial Design	10	26/2/2020
2.2 Roadmap with Release Plan	10	26/2/2020
2.3 Discuss one software development approach which you consider unsuitable for this project	5	4/3/2020
2.4 Use case diagram	10	4/3/2020

Assignment 3

Due dates:-	Hardcopy Blackboard and demo	Various see below
--------------------	------------------------------	-------------------

Design and Prototyping. The design should include the data, architectural, component and interface designs as well as a MarvelApp prototype which will need to be demonstrated. This task ends with a Prototype demonstration

TASK III: Design	Marks	Due date
3.1 Data Design (ERD/class diagram)	10	15/4/2020
3.2 Stand Up		15/4/2020
3.3 Design a storyboard for your proposed solution indicating the most important functionality	10	22/4/2020
3.4 Architectural Design using 4+1 method	10	22/4/2020
3.5 User story 1 – The prototype		
3.5.1 Interface Design (wireframe)	5	“
3.5.2 Prototype Design	5	“
3.5.2 Demonstration using MarvelApp and QR code	15	29/4/2020

Assignment 4

Due dates:-	Hardcopy Blackboard and demo	Various see below
--------------------	------------------------------	-------------------

Development. This section includes the build, test cases and code review for a specific component and ends with a demonstration of software components

TASK IV: Development of a User Story 2 - Use a development language of your choice	Mark	Due date
4.1 Component design	5	6/5/2020
4.2 Test Cases	10	“

	4.3 Peer test of prototype	5	13/5/2020	B
	4.4 Component & Interface Functionality (Demo)	10	“	cla
	4.5 Code Review and Retrospectives (write up)	10	“	B